

Remote Patient Monitoring System

Xuan Li Feng | Imperial College London | 2025

Public technical summary. A desktop application for reviewing live physiological signals, patient records, and alerts through a structured, extensible clinical software architecture.

Engineering objective

How can a desktop system give clinicians a coherent view of multiple patients, live vital signs, abnormal events, and stored clinical information?

System workflow



Implemented functionality

- Multi-patient interface with live ECG and respiration waveform visualisation.
- Patient onboarding, persistent clinical records, and historical review.
- Visual and audio alerts with event acknowledgement and stored alert causes.

Evidence at a glance

CLINICAL WORKFLOW

End to end

From patient onboarding to monitoring, alert acknowledgement, and record review.

System design

- Java Swing client connected to REST services and a PostgreSQL backend.
- Digital Twin visualisation module designed as an extensible interface.
- JUnit test suite and Gradle build workflow supporting quality control.

SYSTEM ARCHITECTURE

3 layers

Presentation, application, and persistent data layers with clear responsibilities.

Project outcome

The completed prototype demonstrated a coherent clinical workflow and a modular software foundation for future monitoring and simulation features, without coupling the interface to one hardware device.